

Generative denoising for template-free gravitational-wave burst search

Panagiotis Minoglou*

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Abstract

Binary black hole (BBH) deep classifiers such as AresGW are optimized for compact-binary morphologies in a fixed mass range. Unmodeled transients (e.g. core-collapse supernovae and other short bursts) lie outside that training objective. We present Keno, a generative residual-search pipeline that learns detector noise, subtracts the predicted noise from whitened strain, and searches the residual with template-free excess power, dual-interferometer coherent lag scans, and an envelope-consistency glitch veto. On unknown-morphology injections in real LIGO noise at 1% false-alarm rate, Keno residual excess power saturates near 100% efficiency across SNR 2–12, while a BBH-trained AresGW-class ResNet plateaus near 60–80% and mismatched matched filtering remains near 20%. On published GWTC/cWB GPS times, coherent residual coincidence recovers 9/18 dual-detector events after vetoing the known L1 glitch contamination of GW170817. Keno is complementary to AresGW: it does not claim superior BBH sensitive distance in the AresGW mass range.

1 Introduction

Matched filtering and BBH-trained deep networks dominate compact-binary searches [Nousi et al., 2023, Koloniari et al., 2025]. AresGW [Nousi et al., 2023] uses a deep residual network with adaptive normalization to classify one-second segments as BBH or noise, achieving competitive sensitive distances in its training mass window. That success does not transfer to morphologies absent from training.

Unmodeled burst searches (e.g. coherent WaveBurst) ask a different question: whether excess coherent power remains after accounting for detector noise. Keno follows that spirit with a generative step: a 1D U-Net predicts instrumental noise N , forming

$$S_{\text{clean}} = S_{\text{raw}} - \hat{N}, \quad (1)$$

then applies template-free excess power on S_{clean} .

Claim scope. We claim superiority on *unknown-morphology* bursts at controlled false-alarm rate. We do *not* claim that Keno supersedes AresGW on BBH detection.

2 Method

Figure 1 summarizes the production path. All strain segments are whitened and sampled at $f_s = 4096$ Hz. Unless noted, analysis windows are $T = 4$ s ($N = 16\,384$ samples).

*Independent researcher. Corresponding author. E-mail: panosmng97@gmail.com

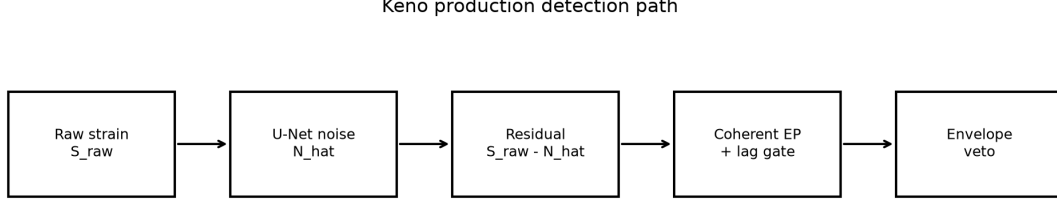


Figure 1: Keno production detection path: generative noise subtraction, residual excess power, coherent H1+L1 lag scan, and envelope-consistency veto.

2.1 Generative noise model

Keno uses a 1D U-Net noise denoiser Ronneberger et al. [2015] with four encoder/decoder stages (base width 16 channels, kernel size 9, GELU activations and batch normalization). The network maps whitened strain S_{raw} to a predicted noise waveform $\hat{N}$ and forms the residual

$$R = S_{\text{raw}} - \hat{N}. \quad (2)$$

Training uses real GWOSC background windows cached for H1/L1. Fine-tuning emphasizes burst preservation so that injected short transients survive subtraction. A Phase-1 hold-out check requires mean normalized recovery error $\text{RMS}(R - s)/\text{RMS}(s) < 0.15$ on injected bursts with known ground truth s . The production checkpoint for this work is frozen under label 2026-07-paper-v1 (SHA256 prefix 55ce7637...).

2.2 Template-free excess-power statistic

On a residual (or raw) time series $x(t)$, Keno computes a short-time Fourier transform (Hann window, $n_{\text{perseg}} = 256$, 75% overlap), forms bin power $P(f, t)$, estimates a robust noise floor as the median of P , and defines the detection statistic as the peak normalized excess power

$$\rho_{\text{EP}}(x) = \max_{f,t} \frac{P(f, t)}{\text{median}(P)}. \quad (3)$$

This is a cWB-style, morphology-agnostic energy excess measure: no template inner product is required.

2.3 Single-detector residual search

For a single interferometer, Keno reports $\rho_{\text{EP}}(R)$ and compares it to a threshold θ_{IFO} calibrated on noise-only windows at a target false-alarm rate (FAR). Calibration uses the empirical $(1 - \text{FAR})$ percentile after trimming the upper 1% of subtraction-glitched noise trials (artifact-robust threshold). In the freeze, $\theta_{\text{IFO}} \approx 4004.6$ at 1% FAR. This gate is used for single-IFO diagnostics and independent residual coincidence; dual-IFO *production* detection uses the coherent threshold below.

2.4 Coherent dual-IFO lag scan

For H1 and L1 residuals R_{H} and R_{L} , Keno scans integer sample lags ℓ with $|\ell|/f_s \leq \Delta t_{\text{lag}}$ ($\Delta t_{\text{lag}} = 10$ ms, the approximate light-travel window between sites) and polarities $p \in \{+1, -1\}$, forming

$$C_{\ell,p} = \frac{R_{\text{H}} + p S_{\ell}(R_{\text{L}})}{\sqrt{2}}, \quad (4)$$

where $\mathcal{S}_\ell$ is a zero-padded shift. The coherent statistic is

$$\rho_{\text{coh}} = \max_{\ell, p} \rho_{\text{EP}}(C_{\ell, p}), \quad (5)$$

with best lag $\tau^* = \ell^*/f_s$.

2.5 Envelope-consistency glitch veto

Independent residual envelope peak times t_H and t_L are estimated from a short smoothed energy envelope (8 ms boxcar), ignoring edge guards of 64 ms so boundary artifacts cannot dominate. Define

$$\Delta t_{\text{env}} = (t_L - t_H) \times 10^3 \quad (\text{ms}). \quad (6)$$

A large $|\Delta t_{\text{env}}|$ with a loud coherent EP is characteristic of single-IFO glitch contamination (e.g. the L1 glitch at GW170817). Production therefore requires

$$|\Delta t_{\text{env}}| \leq \Delta t_{\text{env}}^{\text{max}} \quad (\Delta t_{\text{env}}^{\text{max}} = 50 \text{ ms}). \quad (7)$$

2.6 Production detection rule

A dual-IFO *production* trigger is declared if and only if

$$\rho_{\text{coh}} \geq \theta_{\text{coh}}, \quad (8)$$

$$|\tau^*| \leq \Delta t_{\text{lag}}, \quad (9)$$

$$|\Delta t_{\text{env}}| \leq \Delta t_{\text{env}}^{\text{max}}. \quad (10)$$

The coherent threshold θ_{coh} is calibrated on dual-IFO noise trials that already pass the envelope gate (and, by construction of the lag scan, the timing gate). With 1500 noise trials (23 envelope-gated), the 99th percentile of gated coherent EP yields $\theta_{\text{coh}} \approx 173.1$ at empirical joint FAR 0.07% (1/1500). Critically, $\theta_{\text{coh}} \ll \theta_{\text{IFO}}$: after envelope gating, dual-IFO noise coherent EP is far quieter than the single-IFO residual threshold.

2.7 Baselines (equal FAR)

All baselines share the same whitened windows and are thresholded at the same target FAR on noise-only trials:

- **Oracle matched filter** — upper bound using the true injected waveform (injection studies only).
- **Mismatched matched filter** — fixed sine-Gaussian decoy on raw strain (classical off-morphology failure mode).
- **Raw excess power** — $\rho_{\text{EP}}(S_{\text{raw}})$ without subtraction (cWB-style energy search).
- **AresGW-class** — a compact 1D ResNet trained only on BBH-like chirps versus noise in the same GWOSC cache, then scored as $P(\text{signal})$ on the central 1 s slice. This implements the AresGW *training objective* (BBH vs noise) for off-template evaluation [Nousi et al., 2023]; it is *not* the AUTH published weight file.
- **Keno residual EP** — $\rho_{\text{EP}}(R)$ (production single-IFO path).
- **Keno overlap** — cosine similarity between residual and injected signal (eval-only; requires ground truth).

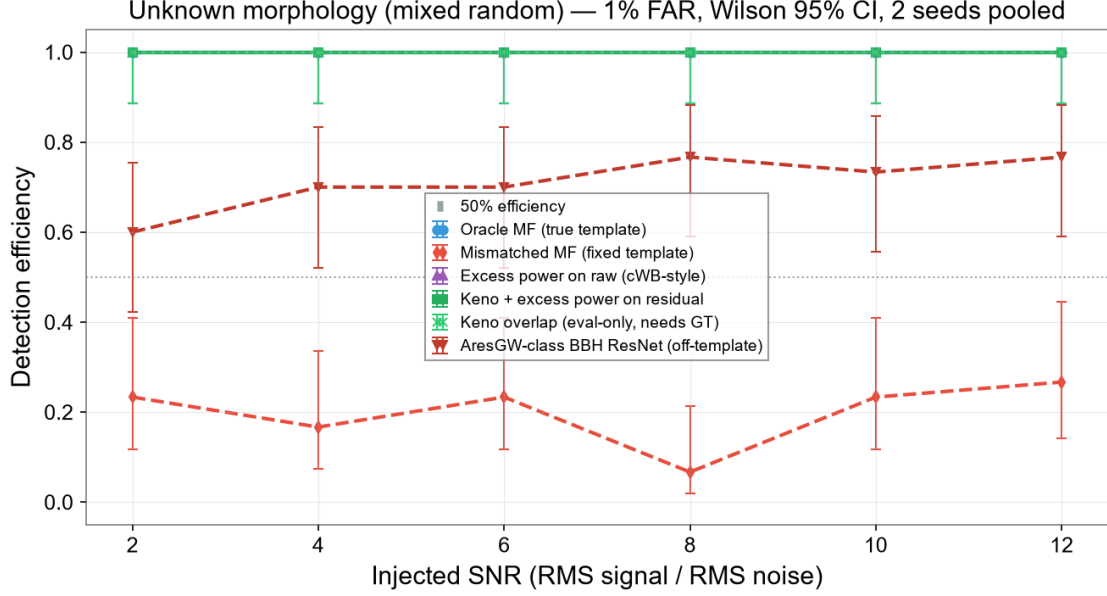


Figure 2: Detection efficiency vs injected SNR at 1% FAR on unknown morphology. AresGW-class is evaluated off-template.

3 Experiments

Injection campaigns use cached real LIGO background with mixed unknown morphologies (sine-Gaussian, ringdown, white-noise burst) at equal FAR. We also report O3 Gravity Spy glitch stress, dual-IFO noise coincidence FAR, and a GWTC/cWB GPS follow-up consistency check (not a discovery claim).

4 Results

4.1 Unknown-morphology efficiency

Figure 2 shows Keno residual EP at 100% efficiency across SNR 2–12 at 1% FAR. The AresGW-class BBH ResNet reaches 50% efficiency but plateaus near 60–80% (off-template leakage without saturation). Mismatched matched filtering remains near $\sim 20\%$ and never reaches 50% efficiency.

4.2 Coherent calibration

Envelope gating removes most dual-IFO noise outliers. With 1500 noise trials (23 envelope-gated), the coherent EP threshold is 173.1 at empirical joint FAR 0.07%, separate from the single-IFO residual threshold ≈ 4005 .

4.3 GWTC/cWB follow-up

Production coincidence recovers 9/18 GWTC/cWB dual-IFO events (Figure 3). GW170817 is rejected by the envelope veto despite huge coherent EP (L1 glitch). Contaminated recoveries are not counted as detections. cWB-only candidates remain 0/3.

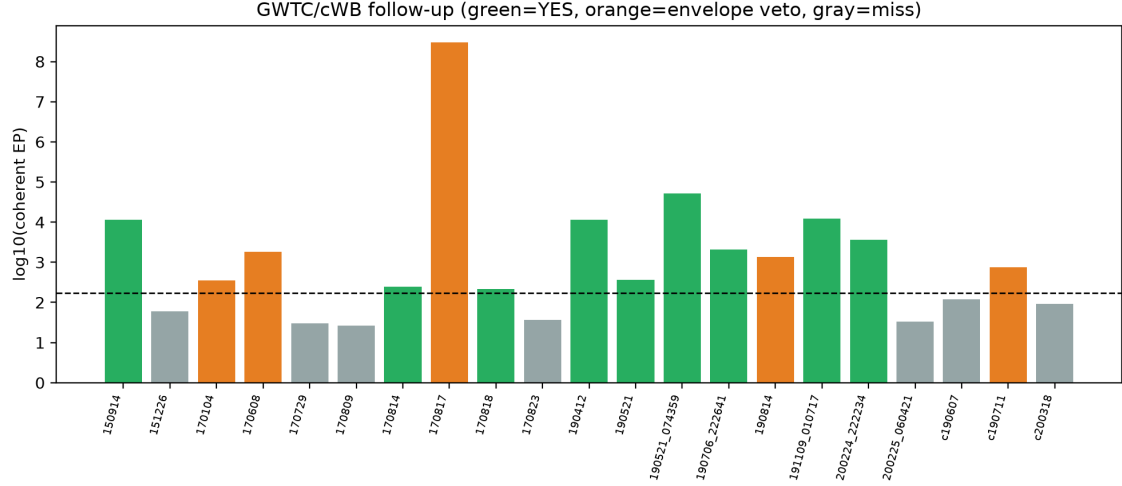


Figure 3: Follow-up coherent EP by event. Green: production YES; orange: envelope veto; gray: miss.

5 Discussion

Keno and AresGW answer different questions. Use AresGW (or matched filtering) for BBH CBC in-band; use Keno when the waveform family is unknown. Limitations include near-zero single-IFO glitch kill rate (coincidence and envelope veto are the defense), residual over-subtraction risk (mitigated by fine-tuning and validation), and incomplete recovery of cWB-only candidates.

6 Reproducibility

Frozen artifacts live under `docs/freeze/current/` with checkpoint SHA256 55ce7637... (label 2026-07-paper-v1). Reproduce with:

```
cd ml-engine
python -m app.evaluation.aresgw_class --train
python -m app.prove --freeze --freeze-label 2026-07-paper-v1
```

7 Conclusion

Generative noise subtraction with template-free coherent residual search recovers unknown-morphology bursts where BBH-trained classifiers fail, at controlled false-alarm rate, without claiming BBH superiority over AresGW.

Data availability

Code and freeze bundle accompany this manuscript. Large evaluation CSVs are hashed in the freeze manifest.

References

P. Nousi et al., Deep residual networks for gravitational wave detection, *Phys. Rev. D* **108**, 024022 (2023).

- A. E. Koloniari et al., New gravitational wave discoveries enabled by machine learning, *Mach. Learn.: Sci. Technol.* (2025).
- O. Ronneberger, P. Fischer, and T. Brox, U-Net: Convolutional networks for biomedical image segmentation, in *Medical Image Computing and Computer-Assisted Intervention (MICCAI)*, pp. 234–241, Springer (2015).